Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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Abstract

Background. Model and preprocessing choices can alter sleep-staging accuracy, computational cost and cross-cohort errors.

Methods. Six configurations, E0 to E4 and E6, were compared on 78 Sleep-EDF subjects using shared 10-fold subject-wise partitions, paired bootstrap and Wilcoxon-Holm analysis. E0 is a convolutional neural network with bidirectional long short-term memory, used as the CNN-BiLSTM baseline. Primary transfer used 180 Sleep Heart Health Study Visit 1 participants, abbreviated SHHS1, without weight updates. Record-wise normalisation in E6 was transductive. Matched interventions included ADAST at limited and full-source budgets.

Results. E3, a one-dimensional residual network with a temporal convolutional network (ResNet-1D-TCN) and fixed-scale preprocessing, reached pooled Sleep-EDF macro-F1 0.7904. It exceeded its record-wise-normalised counterpart E6 by 0.0213, with $CI_{95\%}$ $[0.0122, 0.0307]$ and Holm $p = 0.0012$. A second seed preserved the direction but not Holm significance. E3 ran 3.76× faster than the baseline in the forward pass, with 4.37× as many parameters. On SHHS1, it improved subject-mean macro-F1 over the baseline by 0.0412, with $CI_{95\%}$ $[0.0314, 0.0512]$. Additional same-cohort comparisons favoured band-pass-only E4 over E3 at both seeds, by 0.0131 at seed 42 with $CI_{95\%}$ $[0.0111, 0.0153]$ and by 0.0098 at seed 123 with $CI_{95\%}$ $[0.0073, 0.0125]$. Both primary pipelines misclassified over 72% of deep-sleep N3 epochs as N2; E3 N3 recall was 0.2582. Full-source ADAST improved subject-mean macro-F1 by 0.0445, with $CI_{95\%}$ $[0.0352, 0.0539]$, over its matched control and raised N3 recall from 0.2432 to 0.6111, while lowering N1, N2 and REM F1.

Conclusions. E3 combined faster computation with better aggregate transfer performance than the baseline, whereas band-pass-only E4 achieved higher SHHS1 scores. N3-to-N2 confusion dominated deep-sleep errors. These findings support joint evaluation of model design, preprocessing and stage-specific performance.

Keywords: sleep staging; single-channel EEG; temporal convolutional network; preprocessing; domain shift; paired evaluation.

1. Introduction

Sleep staging assigns one of five labels, namely W, N1, N2, N3 and REM, to every 30 s epoch of a polysomnographic recording. Manual scoring is slow, requires trained expertise, and retains disagreement between scorers; in the inter-scorer reliability programme of the American Academy of Sleep Medicine, N1 shows the lowest agreement of all stages [1]. An automated system therefore has to be judged not by overall accuracy alone, which is dominated by the most frequent stages, but by class-balanced metrics, by stability across individuals, and by its behaviour when the recording device, montage or population changes.

Progress in this field is difficult to compare across published tables. Studies differ simultaneously in dataset version, subject partitioning, wake-trimming policy, filtering, normalisation, sequence length and model-selection rule, making architectural effects difficult to distinguish from other pipeline differences across studies. For practical development, the relevant questions are which pipeline choices improve prediction, what they cost computationally, and whether their benefits persist on another cohort. Answering these questions requires shared evaluation partitions and an explicit account of what changes between configurations, including their training procedures.

We take ZleepAnlystNet [2], a two-stage single-channel model that couples a bank of class-specific convolutional encoders to a BiLSTM, as an internal baseline, E0, reimplemented under the present protocol. E1 replaces the BiLSTM configuration with a temporal convolutional network (TCN) [3], reusing the encoder and cached features but changing the sequence-model training recipe. E2 replaces E1's 75-dimensional current/previous/next feature-context package, abbreviated C/P/N, with a 128-dimensional current-epoch residual encoder [4], also changing encoder training and selection. These are comparisons of specified configurations, not isolated architectural effects. Preprocessing contrasts then examine filtering and amplitude handling within the same ResNet-1D-TCN model and training configuration.

Model and signal-processing choices can change both prediction quality and computational cost, but good performance on one dataset may not carry over to another. We therefore ask three questions:

(i) How do the evaluated model configurations differ in their ability to classify sleep stages and in the computational resources they require?

(ii) With the model architecture and training settings held fixed, how do filtering and amplitude normalisation choices affect classification performance?

(iii) When applied to data from a different group of people without updating model weights, which configurations perform better, and which sleep stages remain difficult to recognise?

Our contributions are the following.

(1) We quantify the trade-offs between classification performance and computational cost across the evaluated configurations. The implemented ResNet-1D-TCN pipeline requires less computation time than the CNN-BiLSTM baseline, despite having more parameters.

(2) We show that preprocessing choices can substantially affect performance on another dataset. Even with the ResNet-1D-TCN architecture and training settings held fixed, filtering and amplitude handling lead to different results on SHHS1. These comparisons show why preprocessing needs to be evaluated alongside the model.

(3) We identify N3-to-N2 confusion as a shared cross-cohort error: both CNN-BiLSTM and ResNet-1D-TCN misclassify more than 72% of reference N3 epochs as N2. Matched intervention experiments quantify how class weighting and domain adaptation increase N3 recall while changing recognition of the other stages.

2. Related work

DeepSleepNet combined a dual-branch CNN with a BiLSTM to capture epoch morphology and temporal dependence [5]. AttnSleep introduced attention over multi-resolution features [6], and SleepTransformer applied self-attention with explicit uncertainty quantification [7]. Together these establish that sequential context matters substantially. They also illustrate the comparability problem: differences in dataset version, subject split and preprocessing between these works are large enough that cross-table comparison cannot substitute for an ablation carried out inside one protocol, a point emphasised in recent reviews of the field [8].

Single-channel systems also span materially different input representations. SleepInceptionNet systematically compared raw, spectral and time-frequency inputs and selected a continuous-wavelet, central-channel Inception model that scores one epoch at a time [9]. That study reinforces two points relevant here: a single EEG channel can support five-stage scoring, and preprocessing choice is part of the model definition rather than a neutral implementation detail.

Cross-dataset sleep staging is already an active methodological problem. For example, He et al. [10] trained a domain-adaptation network across Sleep-EDF, SHHS1 and the PhysioNet Challenge data. ADAST instead uses domain-specific attention, adversarial alignment and iterative pseudo-label self-training on unlabelled target data [11]. These approaches adapt model representations or decision functions using the target domain; our label-free E6 condition only estimates recording-level normalisation statistics, with no learned target-domain or weight adaptation. Van Der Donckt et al. [12] further showed that increasingly complex deep architectures need not outperform a carefully designed conventional pipeline under a matched protocol. Our contribution is a joint assessment of pipeline substitutions, preprocessing, computational cost and cross-cohort error structure. The paired comparisons quantify benefits within a common protocol; the cross-model analysis tests whether an aggregate transfer gain also addresses the dominant stage-specific failure. Neither question is answered by comparing scores from unrelated benchmark tables.

Sleep-EDF Expanded is the standard public benchmark for single-channel EEG [13, 14, 15]. The Sleep Heart Health Study (SHHS) is a large multi-centre cohort assembled to study sleep-disordered breathing and cardiovascular risk [16], distributed through the National Sleep Research Resource (NSRR) [17]. The two differ in age, comorbidity, recording hardware, montage and label distribution. SHHS therefore tests a form of cohort change not represented by a held-out split of Sleep-EDF. We use it to assess cross-cohort robustness, without equating retrospective transfer testing with clinical deployment validation.

Two further strands of work inform the interpretation. Slow oscillations have a spatially varying amplitude and propagate across the cortex [18], while deep-sleep scoring uses amplitude criteria [19, 20]. A recent SHHS analysis examined how those criteria relate to age- and sex-associated differences in N3 scoring [21]. Such work provides context for transfer between derivations and populations, rather than a mechanism established by the present experiment. Differences in class prevalence can also affect predictions; prior correction and probability calibration address related but distinct problems [22, 23]. We therefore examine class-wise errors and label proportions without treating marginal prediction counts as a calibration measure.

3. Materials and methods

3.1. Data and labels

Our Sleep Cassette analysis sample from Sleep-EDF Expanded comprises 153 recordings from 78 subjects; we use the Fpz-Cz derivation at 100 Hz with 30 s epochs (3,000 samples per epoch). Annotations are mapped to W, N1, N2, N3 and REM, merging the original stages 3 and 4 into N3; Movement and Unknown epochs retain their temporal position but are excluded from the loss and from all metrics. After windowing, 195,767 epochs remain, of which 195,469 are scored and 298 are Movement/Unknown.

An evaluation window is applied identically to both cohorts. It is anchored on the first and last true sleep epoch of type N1, N2, N3 or REM, and extended by 30 min of wake on each side, i.e. 60 epochs per side. This policy is stated explicitly because it strongly affects the wake fraction and hence every aggregate metric, and because it is one of the reasons published numbers are not directly comparable. It is a benchmark window, not a deployable inference rule, since its anchors use ground-truth labels.

The SHHS Visit-1 selection used the metadata quality field `overall_shhs1`, admitting values 3-7. Eligible participant identifiers were ranked deterministically by SHA-256 with selection seed 42, then independently ranked for role assignment: 5 adaptation, 15 validation, 180 test and 20 reserve participants. These are the archived protocol's role names; in the primary experiments the adaptation group served technical checks, not model adaptation. No reserve replacements were required. The 180-person test sample was selected before prediction review. Neither the adaptation nor validation group updated weights in the primary experiments; the additional ADAST experiment uses only the five adaptation participants' unlabelled signals for weight updates. The EEG channel is C4-A1 at 125 Hz, resampled to 100 Hz with

Table 1. Scored-label distribution in the two evaluation samples.

Stage	Sleep-EDF		SHHS1 test		
	Epochs	Share	Epochs	Share	Ratio to EDF
W	65,941	33.73%	36,983	21.88%	0.65
N1	21,522	11.01%	7,002	4.14%	0.38
N2	69,132	35.37%	76,287	45.14%	1.28
N3	13,039	6.67%	22,806	13.49%	2.02
REM	25,835	13.22%	25,934	15.34%	1.16
Total scored	**195,469**	**100%**	**169,012**	**100%**	

a polyphase anti-aliasing filter, with up factor 4, down factor 5 and Kaiser window $\beta = 5.0$, applied to the continuous signal before epoching. SHHS labels are mapped 0 →W, 1 →N1, 2 →N2, $\{3, 4\}$ →N3, 5 →REM, $\{6, 9\}$ →excluded. Table 1 reports both label distributions.

Observed class proportions differ, with total variation distance 0.187 and $D_{\mathrm{KL}}(\mathrm{EDF} \,\|\, \mathrm{SHHS}) = 0.101$ nats. SHHS1 contains proportionally 2.02× as much N3 and 0.38× as much N1. We report the per-class ratios in Table 1 to support interpretation of the class-wise errors. These marginal differences alone do not establish the label-shift assumption of unchanged class-conditional inputs.

3.2. Preprocessing conditions

Figure 1 summarises the evaluated architecture, context and preprocessing configurations and their shared evaluation protocol. Calibrated EEG is processed continuously before epoching; the resulting epochs are encoded by either the 15-CNN or ResNet-1D encoder, then modelled as a sequence by a BiLSTM or TCN.

All conditions operate on the continuous calibrated signal in microvolts before epoching. Four active signal conditions enter the performance analysis, and one audit-only condition (E5) records a planned clipping ablation:

- **Raw** (E0-E2): no filtering, no clipping, no rescaling; native μV retained.
- **Band-pass only** (E4): zero-phase Butterworth band-pass, order 4, 0.5-30 Hz, implemented as second-order sections with forward-backward filtering.
- **Band-pass plus clipping** (E5; audit only): the E4 signal followed by a hard clip at $\pm 800\,\mu$V. Across all 153 Sleep-EDF recordings, E5 and E4 were bitwise identical and the measured clipped-sample fraction was zero. Fold 0 was retained as an audit record; the remaining folds and all performance tests were cancelled before scores were inspected because E5 did not define an independent input condition.
- **Fixed-scale amplitude handling** (E3): the same band-pass, followed by a hard clip at $\pm 800\,\mu$V and division by a constant factor of 100. The clip threshold is an absolute constant identical for every recording, so no record-level or cohort-level statistic is estimated and cross-subject leakage is not introduced; relative amplitude between recordings is preserved.
- **Per-record z-score** (E6): the same band-pass and clip, followed by subtraction of the mean and division by the standard deviation computed over the whole filtered, clipped recording. This uses no target labels but does use all samples of the target recording, removes absolute amplitude differences between recordings and is transductive at the recording level rather than purely inductive.

No notch filter is applied: 50 Hz lies outside the 0.5-30 Hz pass-band and at the Nyquist frequency of the 100 Hz signal. The contrast E3-E6 therefore evaluates a practical preprocessing choice: preserving calibrated cross-recording amplitude relationships or applying record-wise normalisation. Because clipping never activated in the Sleep-EDF E5 audit, E3 and E4 differ on Sleep-EDF by the fixed division by 100; the corresponding SHHS audit also found zero clipped samples. The E3-E4 comparison therefore tests training and evaluation at two input scales on these data, not an active clipping effect.

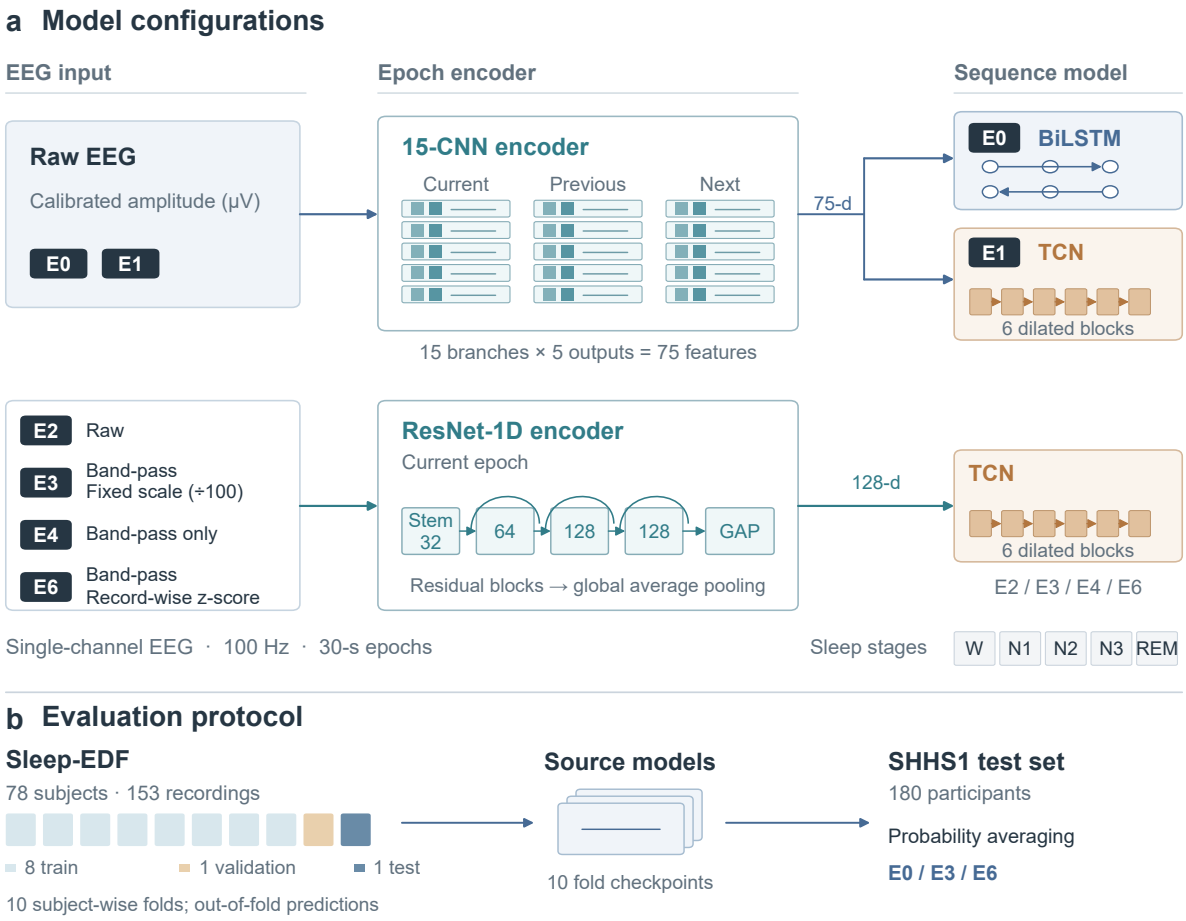


Figure 1. Model configurations and evaluation protocol. (a) E0 and E1 share a 15-CNN encoder: five branches for each of the current, previous and next epochs yield 75 features for a BiLSTM or TCN. E2, E3, E4 and E6 use separately fitted current-epoch ResNet-1D encoders and TCNs. ResNet numbers denote output channels; GAP is global average pooling. Both TCNs have six non-causal dilated blocks. E3 and E6 additionally clip at ±800 μV after band-pass filtering; E3 divides by 100, whereas E6 uses complete-record z-score statistics. (b) Sleep-EDF uses ten shared subject-wise folds, with eight for training, one for validation and one for testing per rotation. The primary SHHS1 evaluation compares E0, E3 and E6 on 180 participants, averaging class probabilities from ten source-fold checkpoints before selecting the predicted class, without target-domain model fitting. E6 uses unlabelled target-record statistics. E5 is an input-identity audit, not an additional performance configuration. Training settings and secondary analyses are described in the Methods and Supplement. The schematic was drawn with Matplotlib and OpenAI Codex assistance; the symbols represent architecture and fold roles, not measured signals or partition sizes.

3.3. Model configurations

Six active configurations, E0-E4 and E6, enter performance analysis; E5 documents the cancelled clipping-only contrast. The baseline encoder concatenates five-class softmax outputs from 15 CNN branches spanning current, previous and next epochs, giving 75 features per central epoch. Context stays within each recording, with edge-epoch replication. ResNet-1D produces 128 features from the current epoch only. The non-causal six-block TCN has a 253-epoch receptive field; the BiLSTM baseline has 128 hidden units per direction. Both sequence models process the retained recording with masked padding. These are offline models: the BiLSTM and TCN use future context, and zero-phase filtering and E6 normalisation use future signal samples. The forward benchmark is not a measure of online decision latency. Layer dimensions and all hyperparameters are reported in Supplementary Methods S1.

Table 2. Experimental configuration codes. The identifiers denote signal-model combinations, not a performance ranking or a sequence of model versions.

ID	Signal condition	Epoch encoder	Sequence model and role
E0	Raw	15-branch CNN	BiLSTM; internal baseline
E1	Raw	15-branch CNN	TCN; sequence-model and training-recipe replacement
E2	Raw	ResNet-1D	TCN; encoder/context and encoder-training replacement
E3	Band-pass + clip + fixed scale	ResNet-1D	TCN; fixed-scale configuration
E4	Band-pass only	ResNet-1D	TCN; band-pass configuration
E5	Band-pass + clip	ResNet-1D	TCN; audit only, identical input to E4
E6	Band-pass + clip + z-score	ResNet-1D	TCN; alternative rescaling

3.4. Two-stage training and subject-wise cross-validation

The 78 subjects are partitioned into 10 subject-disjoint folds. In each evaluation round one fold is the test fold, the next fold (modulo 10) is the validation fold and the remainder are training folds. Both nights of a subject always share the same role. The same partition is used for every configuration, and the test fold never participates in model selection. The training API accepts only training and validation loaders, and encoder fitting and feature caching follow the same subject-fold assignments.

Training has two stages. The epoch encoder uses class weights calculated from training subjects and is selected on validation data: minimum weighted validation loss for each baseline CNN, maximum validation macro-F1 for ResNet. Its selected checkpoint is hash-verified and used to cache features without gradients. The sequence model is fitted on the frozen cache using masked cross-entropy and selected by validation macro-F1. E1 reuses E0 encoder checkpoints but changes the sequence optimiser settings: learning rate 0.01 versus 0.0005, batch size 4 versus 8 recordings, maximum 1,000 versus 300 epochs and patience 10 versus 30 for E0 and E1, respectively. E2 additionally changes encoder optimisation and selection alongside the feature/context replacement. Supplementary Methods S1 provides the complete recipes. The comparisons therefore preserve subject partitions and evaluation, not every training choice.

3.5. Statistical analysis and evaluation protocol

All six configurations completed validation-based selection before the primary seed-42 test predictions were evaluated. In Sleep-EDF, pooled macro-F1 is the descriptive headline metric, but the unit of statistical inference is the subject. Paired subject-cluster bootstrap with 10,000 resamples estimates percentile confidence intervals for the difference in pooled macro-F1 [24], while a two-sided Wilcoxon signed-rank test [25] assesses the subject-level distribution of per-subject differences. Holm correction [26] is applied over the four pre-specified comparisons E1-E0, E2-E1, E3-E2 and E3-E6. Pooled macro-F1 is computed from the sum of subject confusion matrices; subject-mean macro-F1 is the arithmetic mean of individual subjects' scores. They are not interchangeable, and pooled macro-F1 is not a weighted average of individual macro-F1 values. The bootstrap interval describes the pooled-score difference, whereas Wilcoxon evaluates the paired subject differences through their signed ranks. All bootstrap intervals are marginal 95% intervals, not multiplicity-adjusted simultaneous intervals. Supplementary Methods S1 specifies bootstrap seeds and zero-difference handling. Metrics use five fixed classes and assign zero when a class-wise denominator is zero. In the Sleep-EDF analysis, E3-E0 is labelled post-hoc throughout.

We also report win/tie/loss counts, sign dominance $\Delta_{\text{sign}} = (W - L)/(W + L)$ and an unadjusted exact sign test as descriptive robustness checks; they are not part of the pre-specified inferential family.

The completed seed-123 run repeated the evaluation with a second fixed training seed on the same partition. The two seeds are analysed separately; p-values are never pooled, and this repeat provides a sensitivity check across the two initialisations. We report it separately from the primary experiment. The archived

configuration lists seeds 42, 123 and 2025; only the completed runs with seeds 42 and 123 contribute results here.

For SHHS, each configuration uses all ten Sleep-EDF outer-fold checkpoints. Softmax probabilities are averaged in float64 across folds before the final argmax for each epoch. The prespecified primary criterion is subject-mean macro-F1, with E3-E0 and E3-E6 as the two primary comparisons under the same subject-cluster bootstrap and Wilcoxon-Holm treatment, now with bootstrap intervals for the subject-mean difference. The E1/E2 analysis is secondary evidence on an previously evaluated sample, and E3-E2 is explicitly post-hoc. The seed-123 E4 extension has its own four-comparison Holm family, reported in the supplement. A subsequent E4 seed-42 comparison used the restored historical source-selected checkpoints and identical benchmark windows; all ten folds were averaged without target-based ranking. Its paired bootstrap intervals are reported separately, without adding a new p-value to the primary family. Reference-label boundary and stable-region analyses are post-hoc offline diagnostics; their exact masks, exclusions and supports are defined in Supplementary Methods S1.

3.6. Additional intervention experiments

After characterising the SHHS1 errors, we evaluated temperature scaling with EM prior correction, class-weighted TCN training, and a harmonised ADAST comparison [11]. These follow-up experiments use the same 180 SHHS1 participants. Calibration uses all ten seed-123 E3 folds: each temperature is fitted on source validation, and each EM prior on five fixed unlabelled adaptation participants, disjoint from test. Class weighting uses ten matched seed-123 pairs, with the corresponding historical E3 encoder frozen in each fold and a newly trained unweighted control. Both arms average all ten fold probabilities. Training used CPU for folds 0-5 and CUDA for folds 6-9; these timings are excluded from the historical operational benchmark. The earlier fold-0 result is retained separately. ADAST uses ten freshly trained matched CUDA pairs, seed 123, with a source-only dual-head control in each fold. Both arms use 1,140 updates per fold and average all ten fold probabilities; source outer-test results are out-of-fold predictions, not ensembles. The earlier CPU single-fold ADAST pilot is retained separately, not pooled into the CUDA experiment. All arms infer complete target recordings before the same historical scoring mask is applied; calibration therefore uses a rerun full-record E3 control. Model selection uses source validation, and adaptation uses unlabelled target data; target test labels are used for scoring. Supplementary Methods S2 specifies the fitting, selection and adaptation budgets. Paired participant-bootstrap intervals for these experiments are reported separately from the primary tests.

Subsequent development experiments on source fold 0 (seed 123) compared 1,140 with 36,870 updates per model, followed by a fresh full-source ADAST reference and three single-factor loss ablations. All models ran for 30 epochs on T4 CUDA. Checkpoints were selected by maximum source-attention source-validation macro-F1, retaining the first tie; epoch-30 checkpoints were reported separately. The five unlabelled adaptation participants were reused, without source outer-test or SHHS test access. Both attention paths were evaluated on the same source validation set. These development comparisons are separate from the ten-fold transfer evaluation (Supplementary Tables S24 and S25).

A subsequent three-arm check used source fold 1 with the same seed, comparing fresh source-only, ADAST-reference and retained-source-loss models. Each model received 35,820 updates over 30 epochs, covering all 152,825 training epochs per pass. Selection and adaptation data were unchanged in role; the source-validation set contained 23,881 epochs. Both attention paths were evaluated at the selected and epoch-30 checkpoints (Supplementary Tables S28 and S29).

To test transfer-budget sensitivity, we completed ten matched full-source pairs at seed 123: 16 new models on source folds 2-9 and four compatible models reused from folds 0-1, excluding the loss-intervention arms. Each model received 30 complete source-training passes, 35,670-37,620 updates depending on fold, with identical initialisation and source order within each pair. ADAST reused the same five unlabelled adaptation participants. Full-source training also increases target exposure and changes scheduler timing in update units; it is a learning-budget comparison, not an isolated source-coverage intervention. The primary comparison uses epoch 30 in both arms at both budgets. Maximum dual-head logits are softmaxed per fold and probabilities are averaged across ten folds; SHHS uses source attention for source-only and target

attention for ADAST. Sleep-EDF outer-test predictions use the corresponding fold and source attention in both arms. We estimate ADAST minus source-only at each budget and the difference between these effects, using the same 10,000 participant-bootstrap draws across all four systems. These intervals condition on the fitted models and do not include training variability (Table 9; Supplementary Table S30). For the secondary full-source evaluation, each model uses the checkpoint with maximum source-validation macro-F1 through source attention, retaining the first epoch on a tie. Attention paths, aggregation and scoring are unchanged; SHHS labels do not enter selection.

3.7. Operational evaluation

Inference timing was measured on identical Tesla V100 hardware for all configurations and excludes I/O, preprocessing and training. Each measurement used a batch of one synthetic 100-epoch sequence, 20 warm-up passes and three rounds of 100 timed forward passes. Configuration order was shuffled in each round; we report the median over all 300 passes. This benchmark measures the implemented full forward pipeline for one fixed tensor shape, not wall-clock inference on whole-night recordings or algorithmic complexity.

The speed-performance figure pairs seed-42 aggregate scores with these measured forward-pass speed-ups; its two panels distinguish pooled Sleep-EDF scores from subject-mean SHHS1 scores.

4. Results

4.1. Out-of-fold performance on Sleep-EDF

Table 3. Out-of-fold test performance on Sleep-EDF (seed 42, pooled over all folds).

Configuration	macro-F1	**Accuracy**	κ	**F1 (N1)**
E0	0.7754	0.8262	0.7614	0.5009
E1	0.7802	0.8315	0.7683	0.5115
E2	0.7835	0.8341	0.7716	0.5180
E3	**0.7904**	**0.8363**	**0.7754**	**0.5276**
E4	0.7891	0.8360	0.7746	0.5270
E6	0.7691	0.8230	0.7573	0.5124

E3 leads the primary seed-42 experiment on all four metrics (Table 3); the paired results are reported in Table 4.

Table 4. Pre-specified paired comparisons on Sleep-EDF macro-F1 (n = 78 subjects). Holm correction is over the four pre-specified rows; sign-test statistics are unadjusted supplementary robustness checks. E3-E0 is post-hoc and is excluded from the Holm family.

Comparison	**Δ**macro-F1	$CI_{95\%}$	$p_{\mathbf{Holm}}$	**W/T/L**	$\Delta_{\mathbf{sign}}$	$p_{\mathbf{sign}}$
E1-E0	0.004811	[−0.000915, 0.010193]	0.1022	51/0/27	+0.308	0.0088
E2-E1	0.003251	[−0.002370, 0.008808]	0.1036	44/0/34	+0.128	0.3082
E3-E2	0.006962	[0.000305, 0.014520]	0.8989	37/0/41	−0.051	0.7343
E3-E6	**0.021319**	**[0.012179, 0.030698]**	**0.0012**	49/0/29	**+0.256**	**0.0308**
E3-E0 (post-hoc)	0.015024	[0.005746, 0.025871]	n/a	49/0/29	+0.256	0.0308

Among the pre-specified contrasts, only E3-E6 has a positive pooled-metric bootstrap interval, [0.012179, 0.030698], together with a Holm-significant subject-level Wilcoxon result, $p = 0.0012$. The E1-E0 and E2-E1 intervals include zero. E3-E2 has a positive pooled interval, [0.000305, 0.014520], but a non-significant Holm-adjusted Wilcoxon result, $p = 0.8989$, and improved only 37 of 78 subjects, with $\Delta_{\text{sign}} = -0.051$ and $p_{\text{sign}} = 0.734$. This supports a small pooled improvement rather than a consistent subject-level advantage.

The post-hoc E3-E0 contrast adds 0.0150 macro-F1 over E0 with interval [0.005746, 0.025871]; it represents the combined architecture, preprocessing and training configuration.

4.2. Per-class and seed-sensitivity checks

Detailed per-class metrics, the Sleep-EDF confusion matrix and both seed-sensitivity tables are provided in Supplementary Tables S4-S7. In domain, N1 is E3's hardest class and the largest E3-E6 F1 difference is on N3, at +0.0628. All four pre-specified differences remain positive under seed 123, but only E3-E6 has a fully positive bootstrap interval in both runs; its Holm-adjusted result is significant for seed 42 only and its magnitude falls from 0.0213 to 0.0102. The direction is therefore stable across the two fixed seeds, whereas the inferential strength is not.

4.3. Complexity, latency and training cost

Table 5. Full-pipeline forward benchmark on a Tesla V100 (fold 0, seed 42; batch 1, sequence length 100). Throughput is in epochs s^{-1} and peak allocated memory in MiB; I/O and preprocessing are excluded.

ID	Parameters	Median (ms)	Throughput	Peak mem.	Speed-up
E0	248,630	13.4315	7,445	58.81	1.00×
E1	640,950	12.5111	7,993	60.31	1.07×
E2	1,085,578	3.5750	27,972	75.54	3.76×
E3	1,085,578	3.5687	28,021	75.54	3.76×
E6	1,085,578	3.5756	27,967	75.54	3.76×

As Table 5 shows, the ResNet-1D-TCN family provides a measured 3.76× speed-up over the CNN-BiLSTM baseline while carrying 4.37× the parameters. Peak allocated memory is 75.54 MiB versus 58.81 MiB for the baseline, an absolute increase of 16.73 MiB in this benchmark. Figure 2 summarises the measured speed-performance trade-off for the primary seed-42 configurations E0, E3 and E6. On the SHHS1 sample, E3 combines the measured forward-pass speed-up with higher aggregate performance than E0 and E6. The implementation is therefore faster but not smaller. In the forward benchmark, E1 reuses the same encoder as E0 and changes the sequence model, giving a 1.07× speed-up. The larger E2-E0 speed-up accompanies the replacement of 15 independently executed CNN branches with one ResNet-1D encoder in the implemented pipeline.

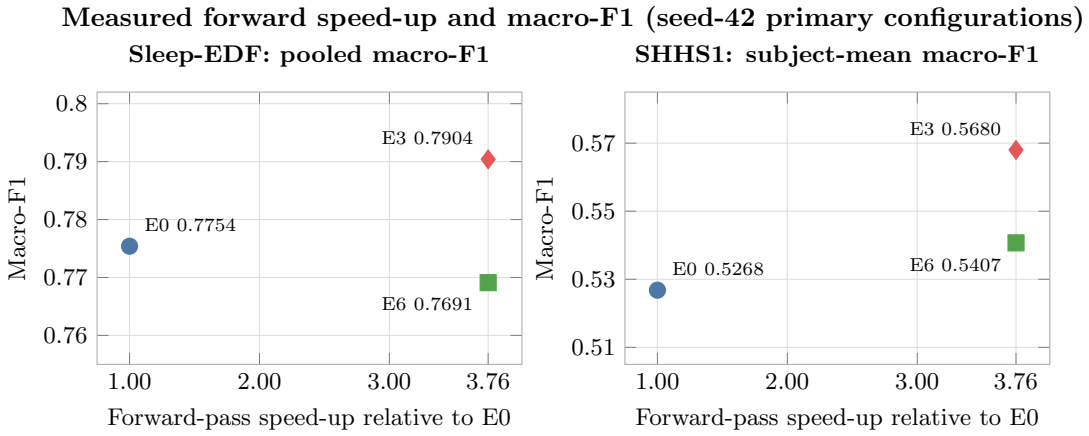


Figure 2. Measured forward speed-up and aggregate macro-F1 **for the primary seed-42 configurations.** Only E0, E3 and E6 are shown. The left panel reports pooled Sleep-EDF macro-F1; the right reports subject-mean macro-F1 on the SHHS1 sample. Speed-up is relative to E0 in the fixed forward-pass benchmark. The plot was generated from archived aggregate results using an editable PGFPlots source; OpenAI Codex assisted with the plotting code.

In the sequential seed-123 run, complete training and validation for E2 and E3 was observed to be 12.2× and 10.2× faster than E0. E1 reused E0 encoder checkpoints and therefore has no end-to-end runtime

comparison. Full timings for the specified training and early-stopping recipes are reported in Supplementary Table S8.

Additional descriptive representation and context-group analyses are reported in Supplementary Fig. S1 and Table S18; they provide complementary evidence on feature geometry and temporal context. In the Sleep-EDF E1 context ablation, adding the explicit Previous and Next (P/N) groups to the current-epoch features produced no detectable boundary-performance gain, with Δmacro-F1 = 0.0010, $CI_{95\%}$ $[-0.0046, 0.0066]$ and Holm $p = 1.000$; 36 of 78 subjects improved. This result is conditional on the E1 encoding and specified TCN recipe.

4.4. Cross-cohort evaluation on SHHS1

The seed-123 run preserves the direction and Holm significance of both SHHS contrasts, whereas the in-domain E3-E6 comparison no longer reaches the Holm threshold. This sensitivity result uses the same SHHS sample; its four-comparison extension family is reported separately in the supplement.

Both primary comparisons support E3 (Table 7), with improvements in 138 and 125 of 180 subjects, respectively. Sign dominance is +0.533 and +0.389; the unadjusted exact sign tests give $p = 3.8 \times 10^{-13}$ and 1.9×10^{-7}. Supporting metrics move together: relative to E0, E3 gains 0.0338 pooled macro-F1, 0.0368 accuracy and 0.0461 κ. Within ± 1 epoch of a reference-label transition, E3's pooled boundary macro-F1 exceeds E0 by 0.0366 and E6 by 0.0183.

The secondary configuration analysis gives a different picture. E1-E0 remains positive but small, at +0.0065 subject-mean macro-F1. E2-E1 is *negative* out of domain, at −0.0128 with an interval entirely below zero, so the bundled replacement of E1's 75-dimensional C/P/N package by the current-epoch 128-dimensional ResNet-1D representation does not generalise better on unprocessed SHHS signals. Because the entire feature/context package changes, the reversal cannot be attributed to the encoder alone. The post-hoc E3-E2 contrast is the largest of the configuration differences reported for this SHHS1 extension: +0.0475, $CI_{95\%}$ $[0.0372, 0.0579]$, $\Delta_{\text{sign}} = +0.633$, with 147 of 180 subjects improved. E3 and E2 use the same model and specified training recipe with different preprocessing. This observed contrast is larger than either the sequence-model/recipe or encoder/context contrast on this sample.

A seed-123 extension evaluating E4, the band-pass-only configuration, provides a more specific secondary contrast. E4 reached subject-mean macro-F1 0.5732, pooled macro-F1 0.6147, accuracy 0.7064, $\kappa = 0.5869$ and N1 F1 0.3492, exceeding same-seed E2 by 0.0323 and E3, at 0.5634, by 0.0098, with paired $CI_{95\%}$ $[0.0073, 0.0125]$ and Holm $p = 1.89 \times 10^{-10}$. The SHHS-specific preprocessing audit recorded zero clipped samples across the 200 processed adaptation, validation and test recordings. E4 and E3 therefore differ on these inputs by the fixed division by 100; each configuration was trained separately on its corresponding source inputs. In this extension, band-pass-only preprocessing outperformed both the raw-input and fixed-scale configurations under the same seed. These comparisons remain secondary because they were conducted after the SHHS sample had been opened. E4 also slightly exceeded E3 on Sleep-EDF at seed 123, with pooled macro-F1 0.7887 versus 0.7883, so this pair does not demonstrate a ranking reversal specific to SHHS1. An additional seed-42 evaluation using all ten historical source-selected E4 checkpoints reached subject-mean macro-F1 0.5811, compared with 0.5680 for E3, a difference of 0.0131 with paired $CI_{95\%}$ $[0.0111, 0.0153]$. Both seed-specific comparisons therefore favour E4 on the same SHHS cohort (Supplementary Table S11); the two seeds are not independent cohorts. The seed-42 comparison uses the original E3 predictions and the same scoring procedure. In this seed-123 extension, E4's pooled N3 recall was 0.2591 and N3 F1 was 0.4068, compared with 0.2410 and 0.3840 for E3. Thus, the higher aggregate score of band-pass-only E4 did not eliminate N3 under-detection. The seed-42 extension showed the same limitation: E4 N3 recall was 0.2728 and N3 F1 was 0.4233, versus 0.2582 and 0.4055 for E3.

Table 6. Cross-cohort performance on 180 SHHS1 participants (seed 42). All model weights were trained on Sleep-EDF. E6 additionally uses complete-record, unlabelled target statistics. Intervals are subject-cluster bootstrap intervals for subject-mean macro-F1.

ID	**Subject-mean** macro-F1 ($CI_{95\%}$)	**Pooled** macro-F1	**Accuracy**	κ
E0	0.5268 [0.5101, 0.5431]	0.5761	0.6648	0.5339
E3	**0.5680 [0.5503, 0.5850]**	**0.6099**	**0.7016**	**0.5801**
E6	0.5407 [0.5227, 0.5584]	0.5732	0.6742	0.5408

Table 7. Primary paired comparisons on SHHS1 subject-mean macro-F1 ($n = 180$).

Comparison	**Role**	Δ	$CI_{95\%}$	p_{Holm}	**W/T/L**	Δ_{sign}
E3-E0	primary	0.0412	[0.0314, 0.0512]	2.65×10^{-13}	138/0/42	+0.533
E3-E6	primary	0.0274	[0.0182, 0.0370]	1.87×10^{-8}	125/0/55	+0.389

4.5. Cross-model errors and their temporal distribution

E3's pooled macro-F1 is 0.7904 on Sleep-EDF and 0.6099 on SHHS1, a difference of 0.1805. Because Sleep-EDF uses out-of-fold predictions whereas SHHS uses ten-model ensembles, the 0.1805 difference is interpreted descriptively. We examine it by class and confusion channel to determine which errors remain despite E3's aggregate transfer advantage. Throughout this analysis, A→B denotes a reference-labelled A epoch predicted as B; it does not denote a physiological stage transition over time.

N3 is the dominant class loss: E3 N3 F1 falls from 0.8085 on Sleep-EDF to 0.4055 on SHHS1, with precision 0.9440 but recall 0.2582. E0 shows nearly the same recall, 0.2610, and N3→N2 rate, 72.3% versus 73.1% for E3. Both model families therefore under-detect N3 despite E3's aggregate transfer advantage, as Table 8 shows. N2→REM is the second-largest E3 confusion channel.

An oracle calculation moves the counts in one selected off-diagonal cell of E3's SHHS confusion matrix to the correct diagonal cell, then recomputes macro-F1. Correcting N3→N2 raises macro-F1 to 0.7444; correcting N2→REM raises it to 0.6596. These changes correspond to 74.5% and 27.5% of the observed pooled-score difference. The calculation uses true labels, and its nonlinear effects are neither additive nor achievable-performance estimates.

The reference-label boundary diagnostic adds temporal detail. For E0 and E3, N3 recall near changes is 0.6121/0.6326 on Sleep-EDF but only 0.0633/0.0733 on SHHS1. Stable-region SHHS recall remains low at 0.4008/0.3900. Thus, N3 under-detection extends beyond the boundary neighbourhoods; it cannot be described solely as an error at label changes. The region definitions and supports are provided in the supplement, including the epochs omitted from the near-versus-stable comparison.

E6 provides a label-free but transductive sensitivity check. The separately trained record-wise z-score configuration has N3 recall 0.2005, compared with 0.2582 for E3; boundary-neighbourhood recall is 0.0721 versus 0.0733. N3 under-detection therefore persists under the tested normalisation configuration. Detailed class-wise, confusion, oracle and transition results are in Supplementary Tables S13-S17.

4.6. Additional intervention results

In the ten-fold calibration experiment, temperature scaling changed subject-mean macro-F1 from 0.5590 to 0.5595, a difference of 0.00049 with $CI_{95\%}$ [0.00002, 0.00097]. Adding EM prior correction reduced it to 0.4899, a difference of −0.0696 from temperature-only with $CI_{95\%}$ [−0.0793, −0.0600]. All ten EM fits converged but placed N3 near the prior floor; the ensemble predicted no N3 epochs (Supplementary Table S19). This full-record comparison uses the same ten models in all three arms, independently of the historical trimmed-context scores.

In the ten-fold matched class-weighting comparison, N3 recall increased from 0.2345 to 0.3225, a

Table 8. N3 under-detection across the primary SHHS1 configurations (seed 42). All models share 22,806 reference N3 epochs. The final column is the proportion of those epochs predicted as N2.

ID	N3 precision	N3 recall	N3 F1	N3→N2 count	Rate
E0	0.9007	0.2610	0.4048	16,480	72.3%
E3	0.9440	0.2582	0.4055	16,674	73.1%
E6	0.9052	0.2005	0.3283	17,764	77.9%

difference of 0.0880 with $CI_{95\%}$ $[0.0805, 0.0954]$, and N3 F1 increased from 0.3759 to 0.4795. However, N2 recall decreased from 0.7260 to 0.6728 and accuracy from 0.6892 to 0.6663. Subject-mean macro-F1 changed from 0.5530 to 0.5514, a difference of −0.0016 with $CI_{95\%}$ $[-0.0049, 0.0017]$; pooled macro-F1 was 0.5930/0.5928 for unweighted/weighted training. Class weighting therefore increased N3 recall and F1 while reducing N2 recall and accuracy; the subject-mean macro-F1 difference interval spans zero (Supplementary Table S20). Within each fold, both arms share the frozen encoder, initialisation and source batch order. This is an intervention on the sequence classifier, not end-to-end encoder retraining; the single-fold pilot is reported separately in Supplementary Table S21.

The limited-budget ten-fold harmonised ADAST comparison improved N3 recall from 0.3195 to 0.4269, a difference of 0.1074 with $CI_{95\%}$ $[0.0714, 0.1422]$ and N3 F1 from 0.4752 to 0.5837. However, subject-mean macro-F1 decreased from 0.4949 to 0.4497, a difference of −0.0452 with $CI_{95\%}$ $[-0.0558, -0.0347]$; pooled macro-F1 decreased from 0.5539 to 0.5072. ADAST produced no N1 predictions on SHHS, compared with N1 recall 0.1974 for the control; REM recall also decreased from 0.4343 to 0.2938. Source out-of-fold pooled macro-F1 fell from 0.6665 to 0.5276, so the deterioration is not restricted to SHHS. N3 gains therefore coexist with substantial other-stage costs and lower aggregate performance. This comparison used seed 123, 1,140 updates per model and five unlabelled adaptation participants (Supplementary Table S22). The earlier CPU single-fold pilot, with source-only/ADAST subject-mean macro-F1 of 0.4755/0.4214, is retained separately in Supplementary Table S23.

At the full-source budget, ADAST increased SHHS1 subject-mean macro-F1 from 0.5010 to 0.5455, a difference of 0.0445 with $CI_{95\%}$ $[0.0352, 0.0539]$, reversing the negative effect at the limited budget. The difference between the two ADAST-minus-control effects was 0.0897 with interval $[0.0767, 0.1023]$, based on the same participant draws across all four systems (Table 9). Pooled macro-F1 increased from 0.5478 to 0.5918. N3 recall increased from 0.2432 to 0.6111 and N3 F1 from 0.3869 to 0.7221; N3→N2 errors decreased from 0.7526 to 0.3689.

Table 9. Learning-budget comparison on 180 SHHS1 participants (169,012 scored epochs). All entries use epoch-30 checkpoints and ten-fold probability ensembles. Source-only is the matched ADAST control, not E3. Full source means 30 complete training passes.

Budget	Arm	Subject-mean macro-F1	Pooled macro-F1	N3 recall	N1 recall
1,140 updates	Source-only	.4949	.5539	.3195	.1974
1,140 updates	ADAST	.4497	.5072	.4269	.0000
Full source	Source-only	.5010	.5478	.2432	.1568
Full source	ADAST	.5455	.5918	.6111	.1057

The full-source gains retain stage-specific costs: N1 recall fell from 0.1568 to 0.1057, and N2 recall from 0.7981 to 0.6369. REM recall increased from 0.6976 to 0.8360, but REM precision decreased from 0.5631 to 0.4375, reducing REM F1 from 0.6231 to 0.5744. Accuracy was 0.6898/0.6873 for source-only/ADAST, a difference of −0.0025 with $CI_{95\%}$ $[-0.0140, 0.0088]$. Full-source ADAST no longer eliminated N1 predictions, but its N1 F1 remained lower than the control. On source outer-test subjects, pooled macro-F1 was 0.7211/0.6831 and subject-mean macro-F1 was 0.6613/0.6281; the latter difference was −0.0332 with interval $[-0.0434, -0.0229]$. Thus, the positive target class-balanced effect coexists with lower source performance. Supplementary Table S30 reports precision, recall and F1 for all five stages on both cohorts.

Selecting checkpoints by source validation retained the positive SHHS1 effect: subject-mean macro-F1 was 0.5073/0.5396 for source-only/ADAST, a difference of 0.0322 with $CI_{95\%}$ [0.0234, 0.0412], and N3 recall was 0.2828/0.6095. N3→N2 errors fell from 0.7110 to 0.3635, while N1, N2 and REM F1 decreased and accuracy was 0.6818/0.6709. Source outer-test subject-mean macro-F1 was 0.6681/0.6368. Thus, the N3 benefit and stage/cohort costs were present under both checkpoint rules, not only at epoch 30 (Supplementary Tables S32 and S33). This is a secondary analysis of the same cohorts and fitted models; epoch 30 remains the primary budget comparison.

On source validation, increasing the learning budget raised selected-checkpoint macro-F1 from 0.6585 to 0.7174 for source-only training and from 0.6470 to 0.6934 for ADAST. With the full-source budget, retaining source-loss weight 1 in round two increased selected-checkpoint N1 recall from 0.2338 to 0.2664, but REM recall decreased from 0.7139 to 0.6490; macro-F1 increased only to 0.6941. Removing adversarial alignment yielded the highest selected macro-F1 among the four ADAST ablations, at 0.6943, with lower N3 recall of 0.7115 versus 0.8261. The no-pseudo-label arm selected epoch 7, before pseudo-label learning starts in the reference; its N1 recall at that checkpoint is not an effect of disabling pseudo-label learning. None of these selected ADAST configurations exceeded the matched full-source source-only macro-F1 of 0.7174. Supplementary Tables S24 and S25 separate selected and final checkpoints and document the stage-wise trade-offs.

The no-adversarial selected checkpoint also performed poorly through target attention on the same source validation set, with macro-F1 0.5227 and N3 recall 0.0773 versus 0.6780/0.8302 for the reference. The highest source-attention score among these ablations therefore coincides with a pronounced N3 loss through target attention (Supplementary Table S27).

On the additional source fold, selected-checkpoint macro-F1 was 0.7370 for source-only, 0.6907 for the ADAST reference and 0.6892 for retained source loss. Retaining source loss again increased selected N1 F1 from 0.2175 to 0.2302, but reduced REM F1 from 0.6852 to 0.6680. N3 F1 increased from 0.8096 to 0.8161 through source attention, whereas it decreased from 0.8003 to 0.7864 through target attention on the same source validation set. Thus, the selected-checkpoint N1 benefit recurred alongside lower REM F1 and aggregate macro-F1. At epoch 30, N1 F1 was lower with retained source loss, at 0.2158 versus 0.2175, showing that the selected-checkpoint improvement did not persist to the final epoch (Supplementary Tables S28 and S29).

5. Discussion

5.1. What the pipeline comparisons establish

The primary comparison shows operational and cross-cohort gains for E3 relative to the CNN-BiLSTM baseline. ResNet-1D-TCN processes the benchmark input 3.76× faster than the CNN-BiLSTM baseline, despite having more parameters. Its observed training-and-validation time is also shorter, although the recipes and stopping rules differ. On the SHHS1 sample, E3 improves subject-mean macro-F1 over E0 by 0.0412, with a positive interval and gains in 138 of 180 participants. These results support E3 as a faster, higher-scoring alternative to that baseline in this setting. For the preprocessing comparison, E4 attained the higher SHHS1 subject-mean score in both seeds.

The intermediate comparisons explain why that benefit should not be credited simply to a newer encoder or sequence model. E1-E0 changes both the sequence model and its training recipe; E2-E1 changes the encoder, feature context and encoder training. Neither contrast reaches the subject-level Holm threshold on Sleep-EDF, and E2-E1 reverses direction on SHHS1. By contrast, the observed post-hoc E3-E2 preprocessing difference is larger on SHHS1. The seed-123 extension further shows that E4, which retains band-pass filtering without fixed scaling, outperforms both E2 and E3 on that sample. Signal handling is therefore a material part of the evaluated model's transfer behaviour, rather than a negligible implementation detail. These configuration-level comparisons do not isolate the contribution of each preprocessing operation in E3-E2.

The E3-E4 contrast is narrower because clipping was inactive in both cohorts. Multiplication by a positive constant preserves waveform shape and relative spectral power; it does not selectively remove slow waves. The

comparison instead concerns separately trained models at two input scales. E4's advantage therefore concerns training and evaluating the model without division by 100, rather than recovering slow-wave information removed by that operation. Separating effects on optimisation from effects on learned normalisation statistics would require additional controlled training comparisons.

5.2. N3 under-detection persists alongside aggregate improvement

The class-wise analysis shows a shared failure mode alongside E3's overall advantage. E0 and E3 both misclassify more than 72% of true N3 as N2, with recall near 0.26 despite precision above 0.90. The alternative E6 pipeline has still lower N3 recall. This common failure persists despite the better aggregate score. The oracle calculation shows the numerical importance of N3→N2 for E3, while the regional analysis shows that the deficit is present in stable N3 as well as around reference-label changes. Together, these observations identify a more specific cross-model error pattern than the overall transfer score alone.

The cohorts differ simultaneously in EEG derivation, population, acquisition and scoring conditions. Slow-wave spatial structure and amplitude-based N3 scoring criteria motivate examining these differences [18, 21]. The present design measures their combined effect rather than separating individual causes. E6 evaluates record-wise z-scoring during both source training and target inference; its lower N3 recall shows that this particular normalisation pipeline further reduces N3 detection.

The intervention experiments show different responses to attempts to increase N3 detection. EM converged to priors near zero for N3 and eliminated N3 predictions. Class weighting increased N3 recall and F1 but reduced N2 recall and accuracy, with a subject-mean macro-F1 difference of -0.0016 with $\mathrm{CI}_{95\%}$ $[-0.0049, 0.0017]$. Limited-budget ADAST increased N3 recall but reduced aggregate macro-F1 and eliminated target N1 predictions. With matched full-source training, ADAST instead improved target subject-mean macro-F1 by 0.0445, with a 0.3679 increase in N3 recall; N1, N2 and REM F1 remained lower than their matched controls. The sign change in the adaptation effect shows why training budget matters when interpreting a UDA baseline. Greater REM recall did not imply better REM F1 because precision fell. Thus, an increase in N3 recall should be assessed together with precision, other-stage errors and aggregate performance. N2→REM provides a second important error channel for E3.

5.3. Positioning and limits of generalisation

This study connects paired pipeline comparisons with computational trade-offs and cross-cohort error structure, complementing work on preprocessing, input representations and conventional/deep pipeline comparisons [9, 12]. E0 is a reimplementation of the ZleepAnlystNet design under the stated protocol. The primary comparisons evaluate transfer without weight updates; the additional ADAST experiment evaluates target-domain weight updates against a matched source-only control.

The study evaluates transfer from Sleep-EDF to SHHS1, with two training seeds for the principal configuration comparisons. Follow-up interventions reuse the same SHHS1 test cohort; adaptation uses five separate unlabelled participants. Sleep-EDF uses out-of-fold predictions and SHHS uses ten-model ensembles, so their score difference combines cohort and evaluation-procedure differences. The models use future context and were evaluated offline; forward-pass timing therefore describes offline processing rather than real-time deployment. Class-wise results are pooled across participants, without sex- or gender-stratified analysis. Further evaluation across cohorts, montages and subgroups would test the consistency of the observed N3 pattern and intervention trade-offs.

6. Conclusion

The evaluated ResNet-1D-TCN pipeline combined lower computational time with improved aggregate performance relative to the CNN-BiLSTM baseline. E3 achieved pooled Sleep-EDF macro-F1 0.7904 and improved subject-mean macro-F1 on the 180-person SHHS1 sample by 0.0412 over E0, with Holm-adjusted significance. Its forward pass was 3.76× faster in the fixed benchmark, and its seed-123 training-and-validation time was 10.2× shorter, despite the larger parameter count. These are benefits of the evaluated configurations, including their training recipes.

Preprocessing materially affected the observed cross-cohort ranking. The seed-42 E3-E2 post-hoc contrast

favoured the combined filtering/clip/fixed-scale pipeline over raw inputs, whereas additional comparisons at seeds 42 and 123 favoured band-pass-only E4 over same-seed E3. E4 is therefore the stronger preprocessing option for SHHS1 among these same-seed comparisons. The results support evaluating signal processing jointly with model design.

E0 and E3 both classified more than 72% of SHHS1 N3 epochs as N2, and N3 under-detection persisted in the tested record-wise normalisation configuration. EM prior correction did not remedy this deficit. Additional ten-fold class-weighted training improved N3 recall at the cost of N2 recall and accuracy, with a subject-mean macro-F1 difference interval spanning zero. Ten-fold ADAST's effect depended on learning budget: the limited-budget comparison reduced class-balanced performance and eliminated N1 predictions, whereas full-source training increased subject-mean macro-F1 by 0.0445 and N3 recall from 0.2432 to 0.6111 relative to its matched source-only control. The full-source benefit retained costs in N1, N2 and REM F1 and source outer-test performance. Together, these experiments identify N3-to-N2 confusion as a principal transfer error and quantify the gains and costs of three intervention strategies. Model and preprocessing selection should therefore combine computational measurements with aggregate and stage-specific performance.

Reproducibility and data availability

Code, protocol snapshots, data partitions and aggregate analyses are retained in the project repository at `https://github.com/quanuiter/SleepTCN`. The repository does not contain restricted participant-level SHHS data, identifiers or predictions. Sleep-EDF Expanded is available from PhysioNet (`https://physionet.org/content/sleep-edfx/1.0.0/`); SHHS access and applicable use conditions are managed by NSRR (`https://sleepdata.org/datasets/shhs`). Participant-level SHHS records and predictions are not included in the manuscript package because access to the source records is controlled by the data provider. The historical snapshot `configs/shhs_zero_shot_v1.json` matches the protocol hash in the original SHHS run manifest. The primary experiment's 540 ensemble prediction files were checked against their recorded SHA-256 values, and their confusion counts reproduce the reported pooled results. The expanded post-run protocol record is preserved separately. Further provenance details are given in the supplement and the repository artifact index.

An archived repository snapshot is available at Git commit `f7c22e7`. Experiment-specific configurations and manifests accompany the retained source history and identify the inputs and settings for each analysis.

Ethics

This study is a secondary analysis of de-identified repository data; no new participants were recruited. The authors confirm that access to SHHS for this analysis was obtained through the National Sleep Research Resource and that no ethics approval was required for this secondary analysis of de-identified repository data.

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CRediT authorship contribution statement

Ngo Nhat Quan: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing - original draft, Writing - review and editing. Pham Thai Son: Validation, Investigation, Visualization, Writing - review and editing. Tri Nguyen Ho Duy: Supervision, Writing - review and editing.

Declaration of competing interests

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

OpenAI Codex assisted with language editing, manuscript organisation, code and result consistency checks, reproducible plotting, and preparation of the vector study-design schematic. It also assisted with implementing and executing the additional analysis, inference and training scripts and their verification tests. Numerical results were computed by those scripts from recorded datasets and model checkpoints, not supplied as synthetic observations by a language model. Historical predictions were retained separately from the additional experiments. The authors remain responsible for reviewing the implementation, experimental design, results and submitted content, including this disclosure.

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